Sales and Distribution (SD)

Curriculum: Introduction to S/4HANA using Global Bike



SAP University Alliances

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Teaching material - Information

Teaching material - Version

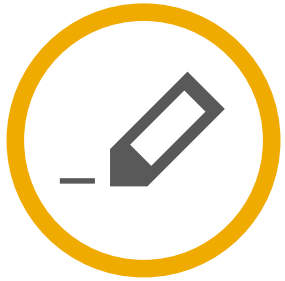
- 4.3 (July 2025)

- Software used
  - S/4HANA 2023
  - Fiori 3.0
- Model
  - Global Bike

- Prerequisites
  - No Prerequisites needed

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Module Information



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- Babett Ruß



Target Audience

- Beginner

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Module Information



Learning Objectives

You are able to

- name some functionalities of the SD module.
- define the central organizational structures of the SD module.
- summarize the master data, which is most important for the SD module.
- explain a standard sales process.

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Functionality

- Sales Support
- Sales
- Shipping and Transportation
- Billing
- Credit Management
- Foreign Trade

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Unit Overview

- SD Organizational Structure
- SD Master Data
- SD Processes
  - Order-to-Cash Process

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SD Organizational Structure

- Client
  - An independent environment in the system
- Company Code
  - Smallest organizational unit for which you can maintain a legal set of books
- Credit Control Area
  - An organizational entity which grants and monitors a credit limit for customers.
  - It can include one or more company codes
- Sales Organization
  - An organizational unit responsible for the sale of certain products or services.
  - The responsibility of a sales organization may include legal liability for products and customer claims
- Distribution Channel
  - The way in which products or services reach the customer
  - Typical examples of distribution channels are wholesale, retail, or direct sales

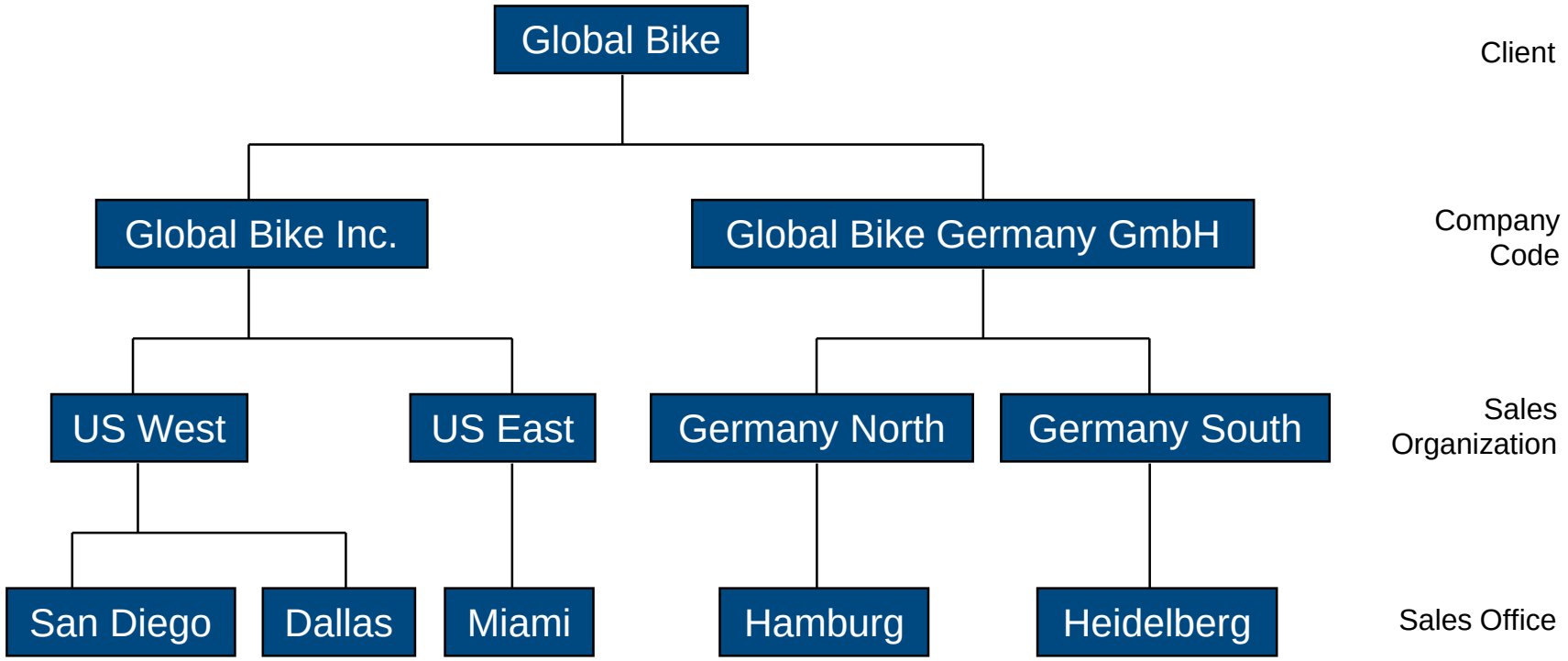
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SD Organizational Structure

- Division
  - A way of grouping materials, products, or services
- Sales Area
  - Combination of Sales Organization, Distribution Channel and Division
  - Determines conditions (i.e. pricing) for sales activities
- Plant (here: Delivering Plant)
  - Plant from which the goods should be delivered to the customer
- Other SD organization units:
  - Shipping Point
  - Loading Point
  - Sales Office
  - Sales Group
  - Sales Person

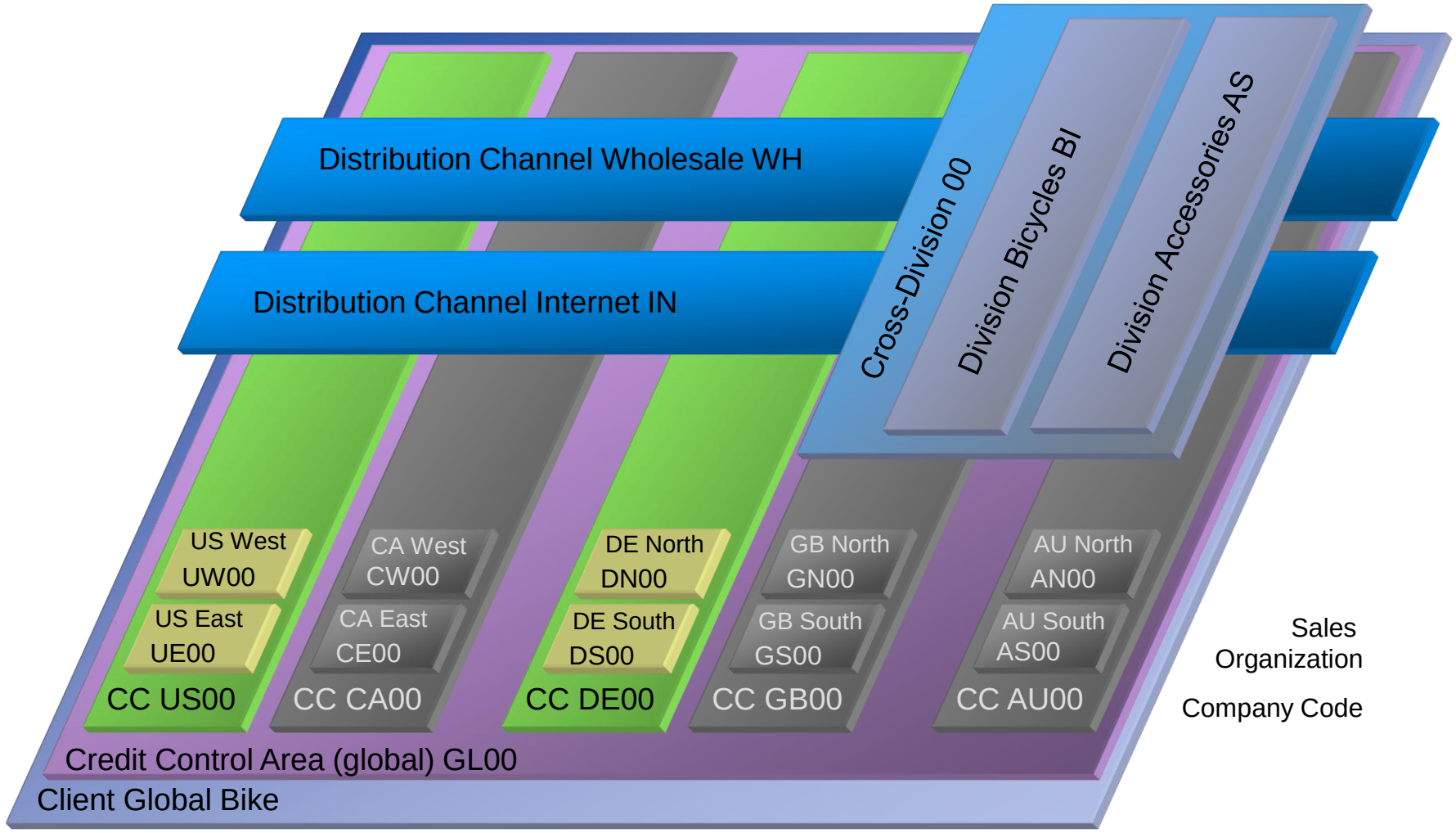
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Global Bike Structure for Sales and Distribution



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Global Bike Enterprise Structure in SAP ERP (Sales)



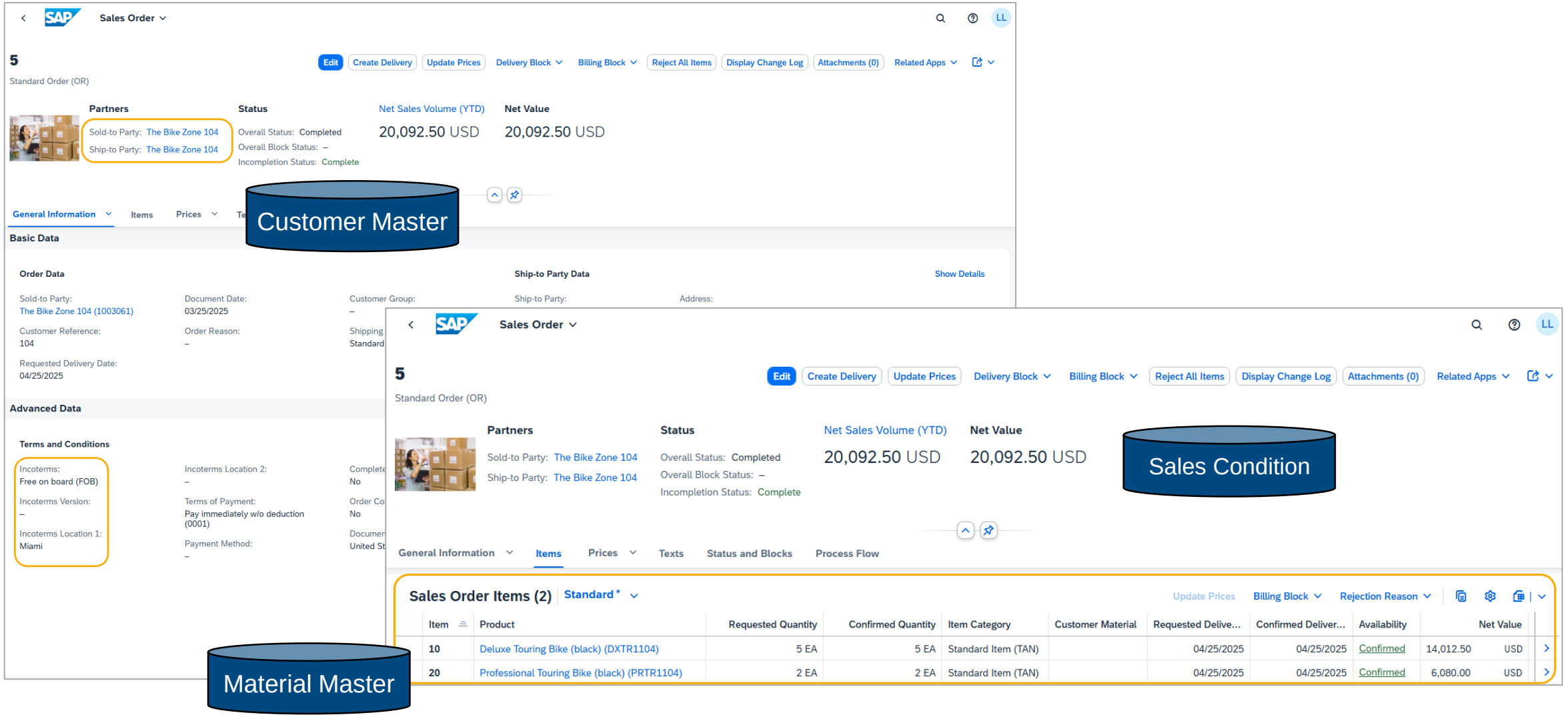
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Unit Overview

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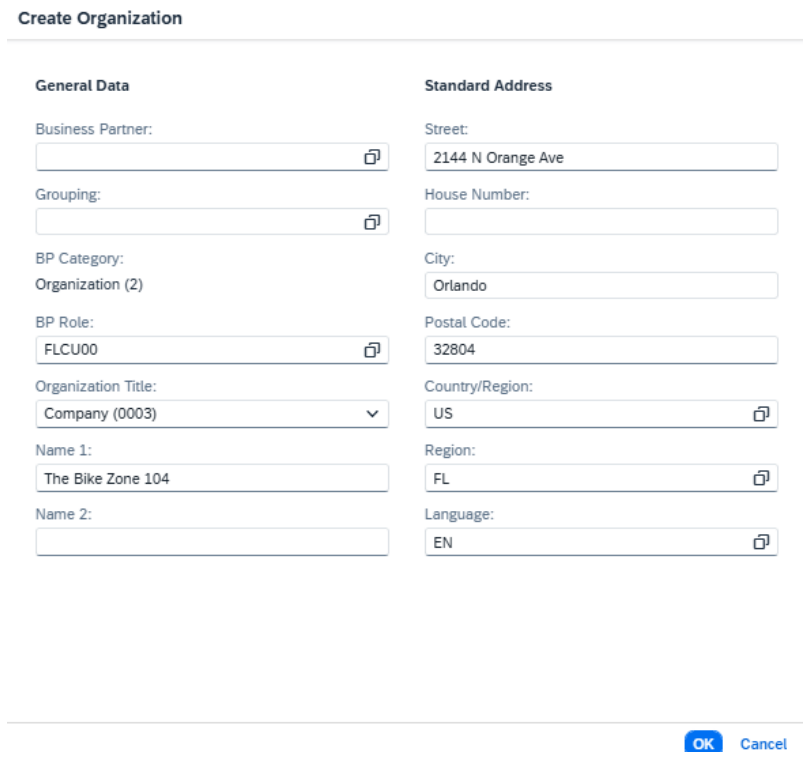
SD Master Data



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Customer Master Data

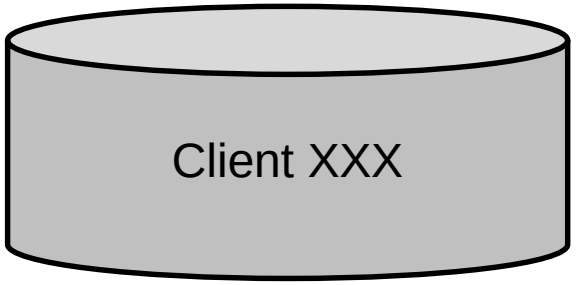
- Customer Master
  - Contains all of the information necessary for processing orders, deliveries, invoices and customer payment
  - Every customer MUST have a master record
    - But you can have one-time customers
- Created by Sales Area
  - Sales Organization
  - Distribution Channel
  - Division
- The customer master information is divided into 3 areas:
  - General Data
  - Company Code Data
  - Sales Area Data



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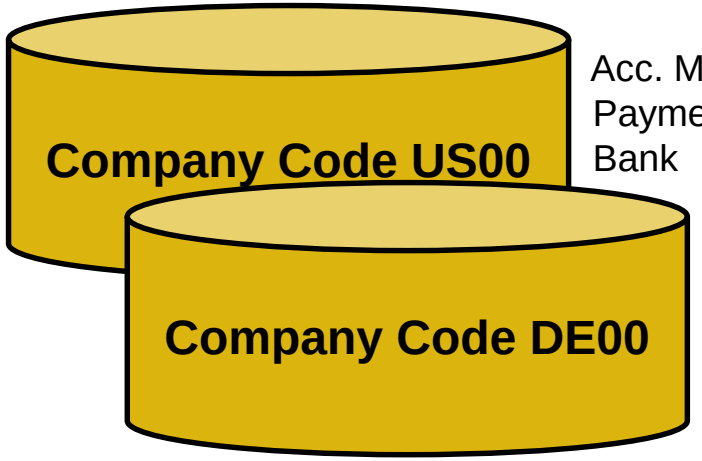
Customer Master

General Information relevant for the entire organization:



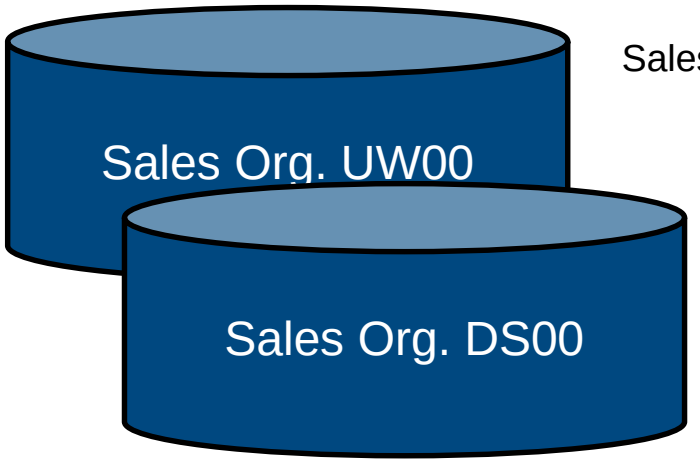
Name
Address
Communication

Company Code specific information:



Acc. Mgmt
Payment
Bank

Sales Area specific information:



Sales Office Currency

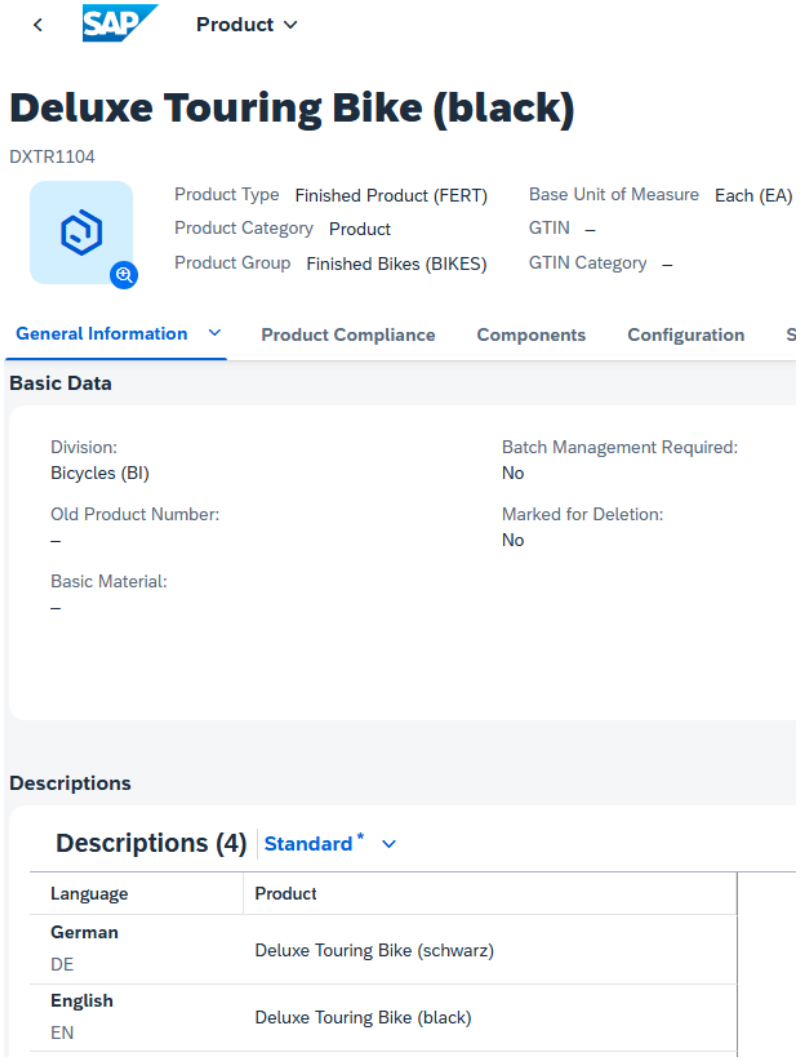
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Material Master Data

- Material Master contains all the information a company needs to manage about a material

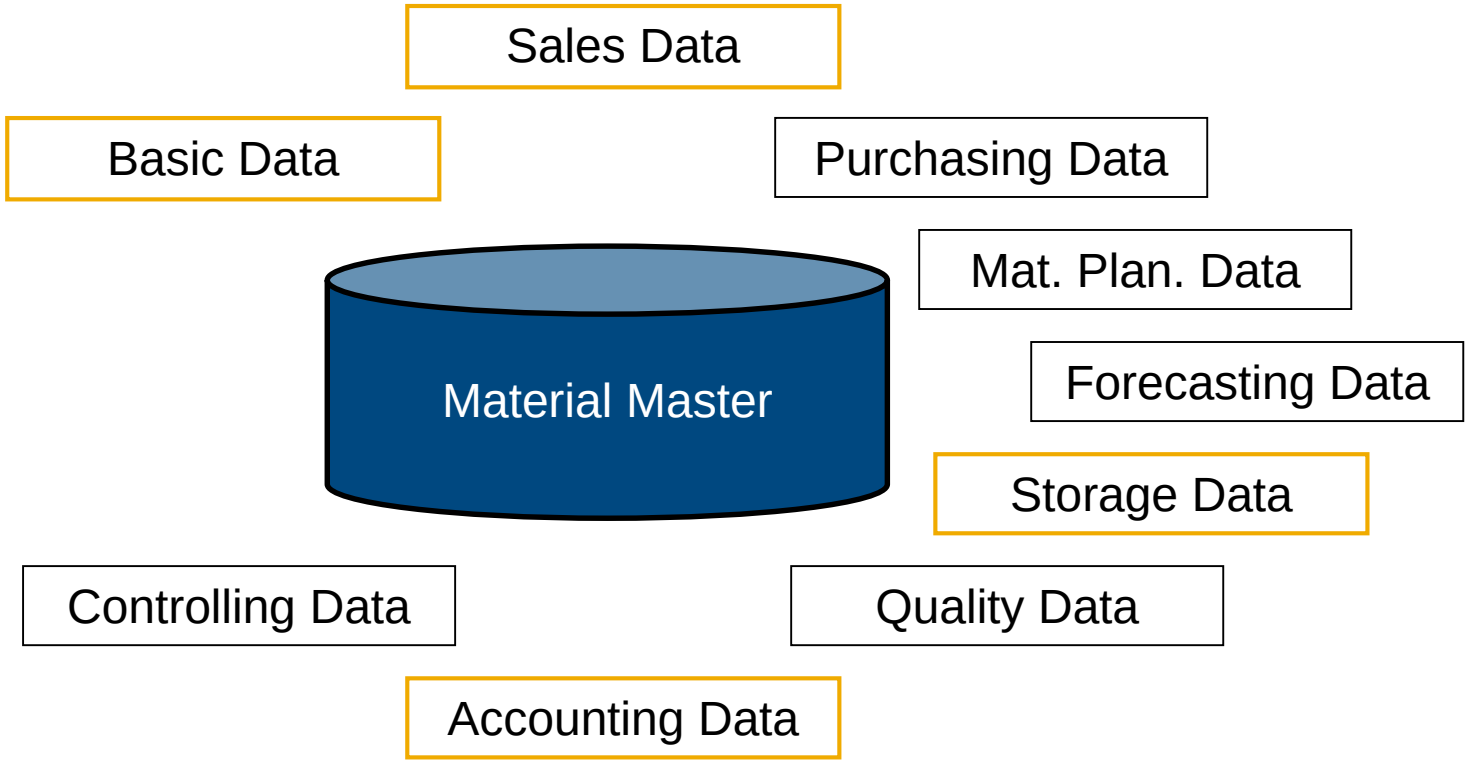
- It is used by most components within the SAP system
  - Sales and Distribution
  - Materials Management
  - Production
  - Plant Maintenance
  - Accounting/Controlling
  - Quality Management

- Material master data is stored in functional segments called Views



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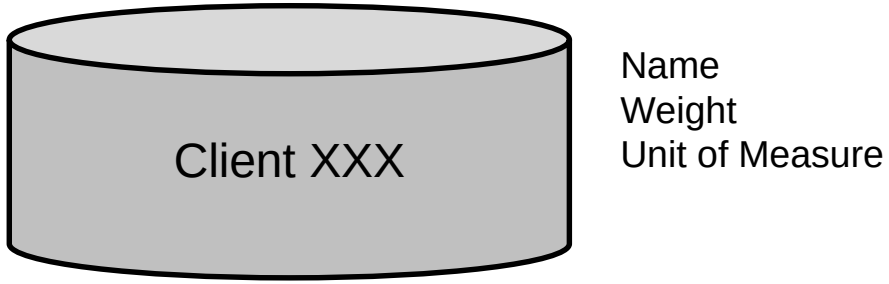
Material Master Views



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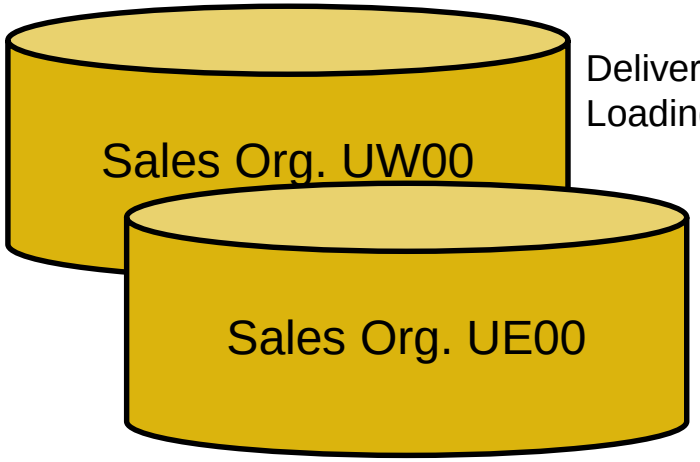
Material Master

General Information relevant for the entire organization:



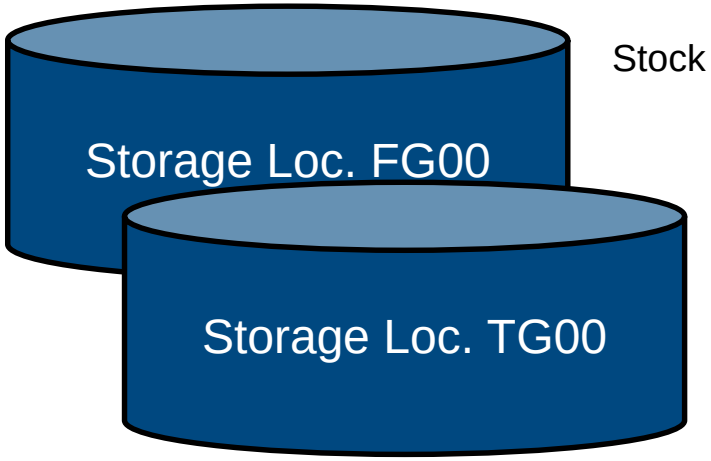
Name
Weight
Unit of Measure

Sales specific information:



Delivering Plant
Loading Grp

Storage Location specific information:

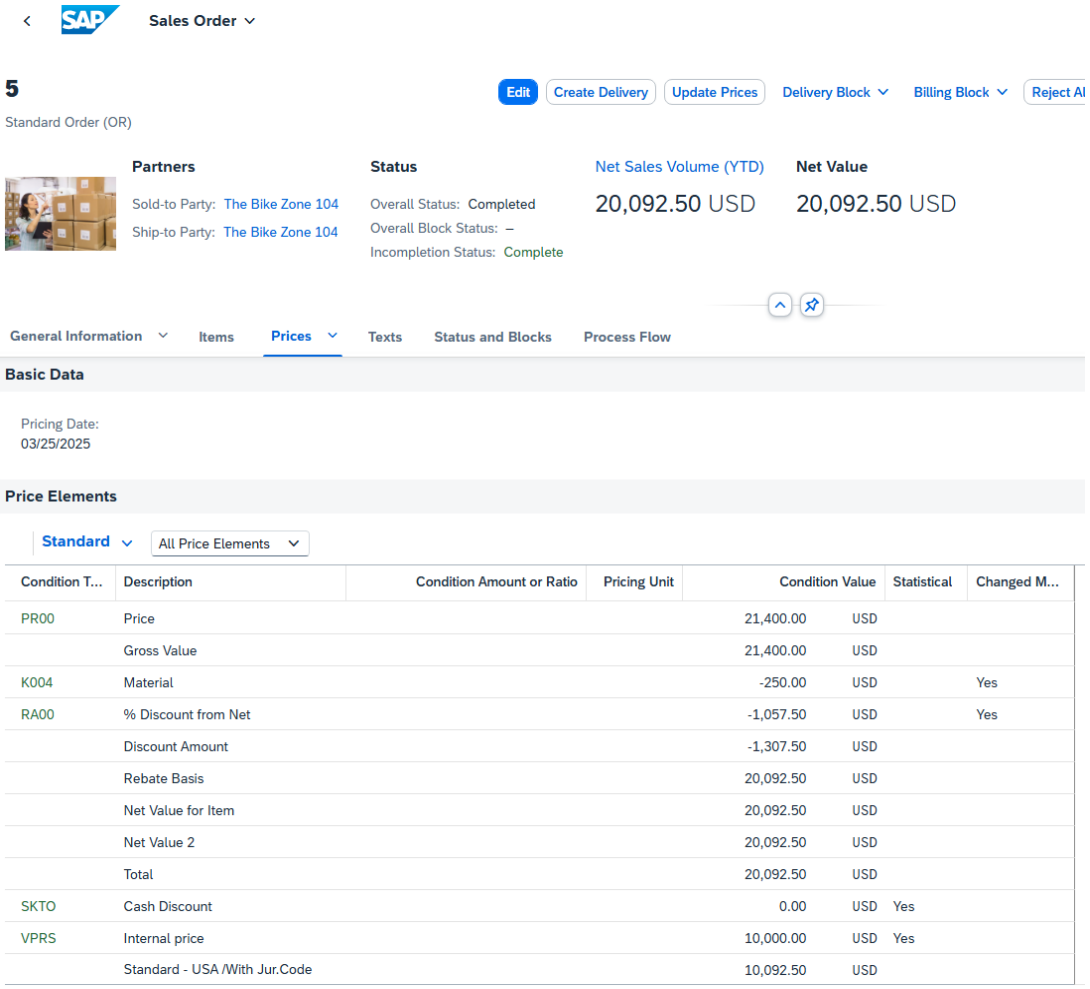


Stock Qty

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Condition Master Data (Pricing)

- Condition master data includes:
  - Prices
  - Surcharges
  - Discounts
  - Freights
  - Taxes
- You can define the condition master to be dependent on various data:
  - Material specific
  - Customer specific
- Conditions can be dependent on any document field



Condition T...	Description	Condition Amount or Ratio	Pricing Unit	Condition Value		Statistical	Changed M...
PR00	Price			21,400.00	USD		
	Gross Value			21,400.00	USD		
K004	Material			-250.00	USD		Yes
RA00	% Discount from Net			-1,057.50	USD		Yes
	Discount Amount			-1,307.50	USD		
	Rebate Basis			20,092.50	USD		
	Net Value for Item			20,092.50	USD		
	Net Value 2			20,092.50	USD		
	Total			20,092.50	USD		
SKTO	Cash Discount			0.00	USD	Yes	
VPRS	Internal price			10,000.00	USD	Yes	
	Standard - USA /With Jur.Code			10,092.50	USD		

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Output

- Output is information that is sent to the customer using various media, such as:
  - Electronic Data Interchange (EDI)
  - Email
  - Mail
  - Fax
  - XML
- Functions for sales, shipping, transportation, and billing
- Examples of output types in SD processing are order confirmations, freight lists, and invoices
  - The output type defines how the output should be transmitted (sent via EDI, or as a print-out)
- Generation of
  - Sales activity output (e.g. customer telephone calls, mailing campaigns)
  - Group output (e.g. freight lists)

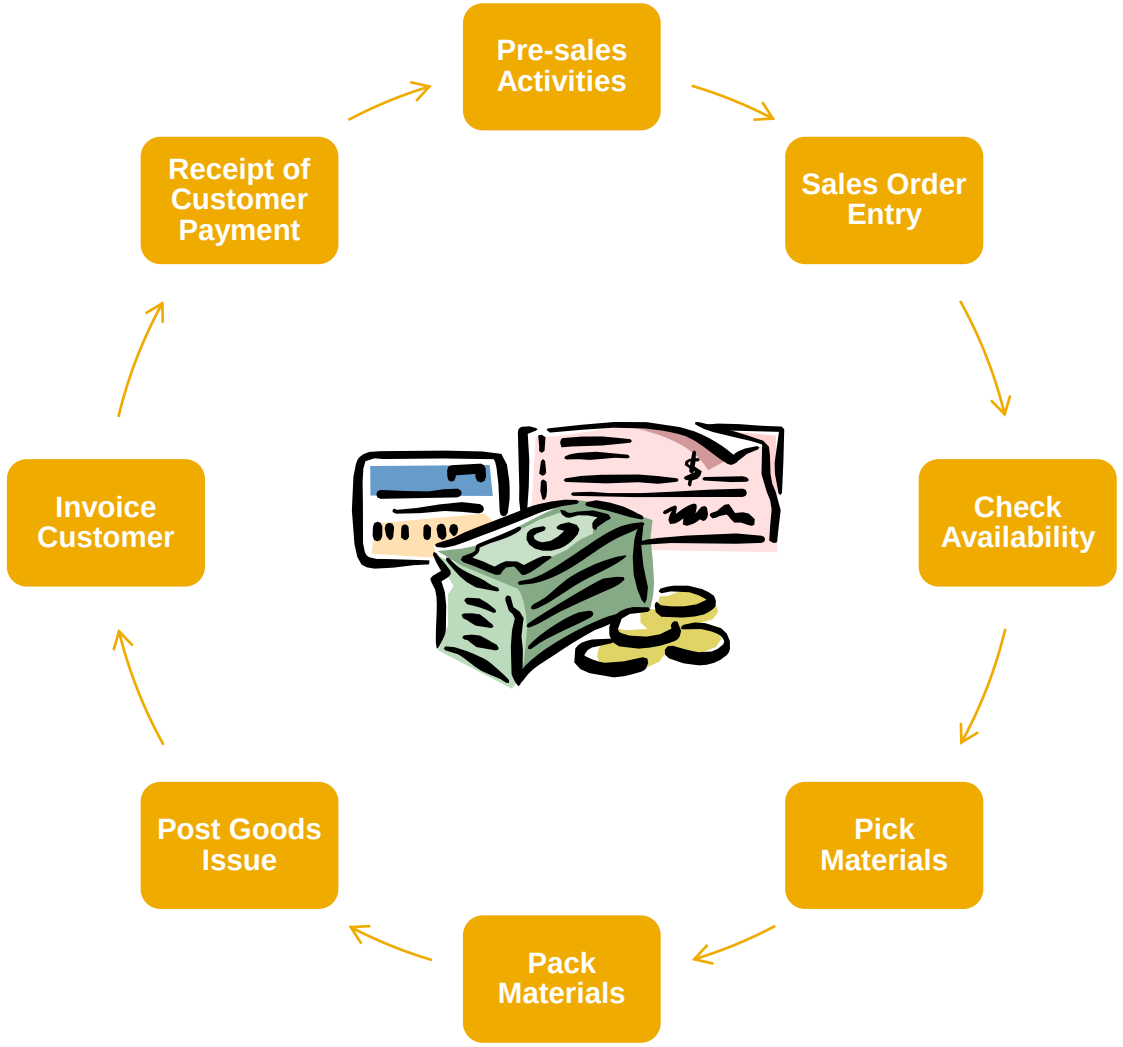
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Unit Overview

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Sales Order Process



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Pre-Sales Activities (CRM Light)

- **Sales Support** is a component of SD that assists the processes of sales, distribution, and marketing of companies as well as their product and services to its customers

- It contains the following functionality
  - Creating and tracking customer contacts and communications (sales activity)
  - Implementing and tracking direct mailing, internet, and trade fair campaigns based on customer attributes

- Pre-sales documents: Inquiries and Quotations

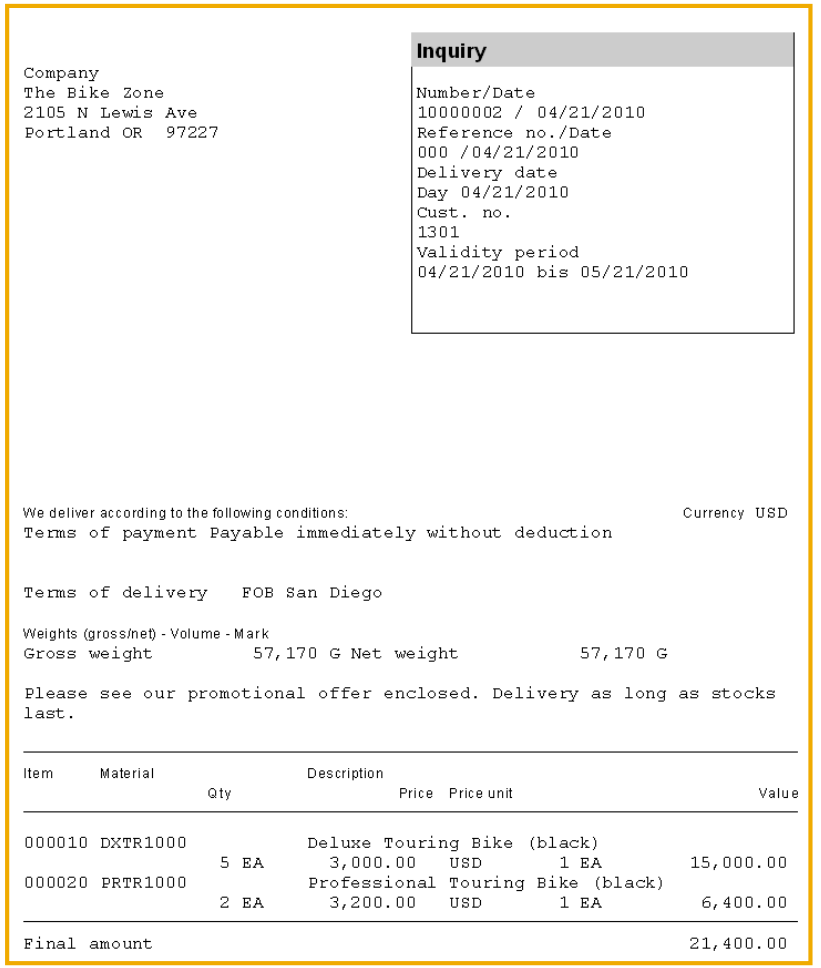
- The ultimate goal of all pre-sales activities is to equip the sales technician with all the **information** necessary to negotiate and complete the potential sale
  - Past sales activity
  - Past communication
  - Contact information
  - General company info
  - Credit limits and usage
  - Current backorders

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Inquiry

- An inquiry is a customer’s request to a company for information or quotation in respect to their products or services without any obligation to purchase
  - How much will it cost
  - Material/Service availability
  - May contain specific quantities and dates

- The inquiry is maintained in the system and a quotation is created to address questions for the potential customer.



Company
The Bike Zone
2105 N Lewis Ave
Portland OR 97227

Inquiry

Number/Date
10000002 / 04/21/2010
Reference no./Date
000 /04/21/2010
Delivery date
Day 04/21/2010
Cust. no.
1301
Validity period
04/21/2010 bis 05/21/2010

We deliver according to the following conditions: Currency USD
Terms of payment Payable immediately without deduction

Terms of delivery FOB San Diego

Weights (gross/net) - Volume - Mark
Gross weight 57,170 G Net weight 57,170 G

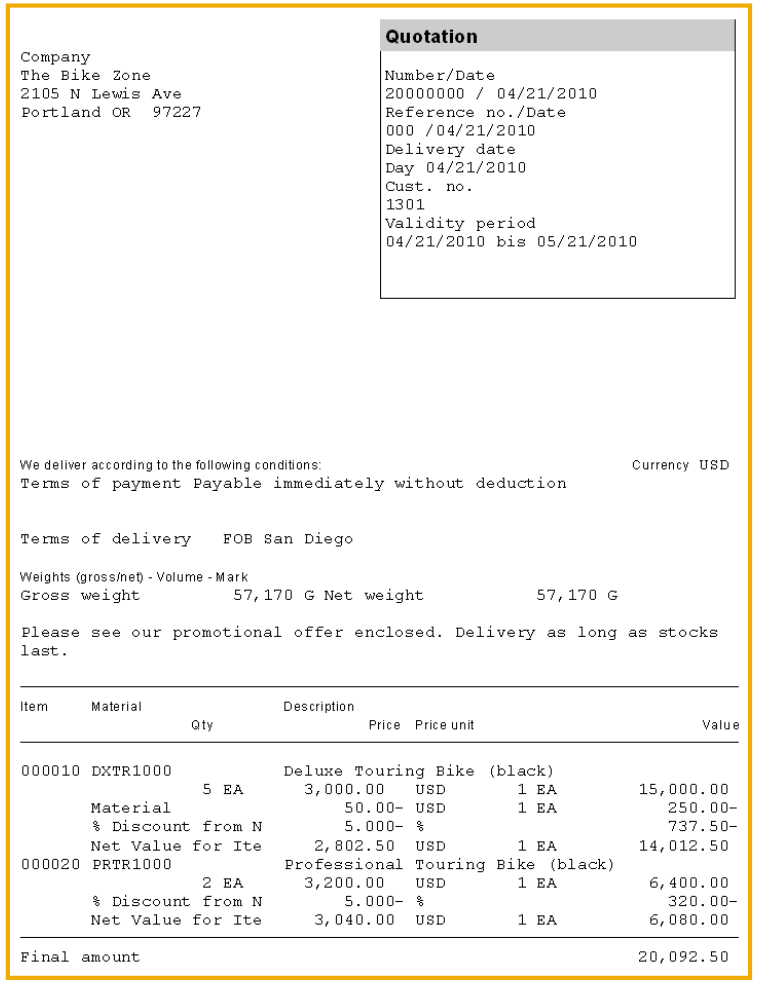
Please see our promotional offer enclosed. Delivery as long as stocks last.

Item	Material	Qty	Description	Price	Price unit	Value
000010	DXTR1000	5 EA	Deluxe Touring Bike (black)	3,000.00	USD 1 EA	15,000.00
000020	PRTR1000	2 EA	Professional Touring Bike (black)	3,200.00	USD 1 EA	6,400.00
Final amount						21,400.00

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Quotation

- The quotation presents the customer with a legally binding offer
  - to deliver specific products or a selection of a certain amount of products
  - in a specified timeframe
  - at a pre-defined price.
- Legally binding on the vendor
- Conditions can apply at various levels:
  - to the entire quotation
  - at item level, to the material to be supplied or to the planned procurement in the case of services
  - at service-line level for individual services (tasks or activities)



Company
The Bike Zone
2105 N Lewis Ave
Portland OR 97227

Quotation

Number/Date
20000000 / 04/21/2010
Reference no./Date
000 /04/21/2010
Delivery date
Day 04/21/2010
Cust. no.
1301
Validity period
04/21/2010 bis 05/21/2010

We deliver according to the following conditions: Currency USD
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Terms of delivery FOB San Diego

Weights (gross/net) - Volume - Mark
Gross weight 57,170 G Net weight 57,170 G

Please see our promotional offer enclosed. Delivery as long as stocks last.

Item	Material	Description	Qty	Price	Price unit		Value
000010	DXTR1000	Deluxe Touring Bike (black)					
			5 EA	3,000.00	USD	1 EA	15,000.00
	Material			50.00-	USD	1 EA	250.00-
	% Discount from N			5.000-	%		737.50-
	Net Value for Ite			2,802.50	USD	1 EA	14,012.50
000020	PRTR1000	Professional Touring Bike (black)					
			2 EA	3,200.00	USD	1 EA	6,400.00
	% Discount from N			5.000-	%		320.00-
	Net Value for Ite			3,040.00	USD	1 EA	6,080.00
Final amount							20,092.50

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Sales Order

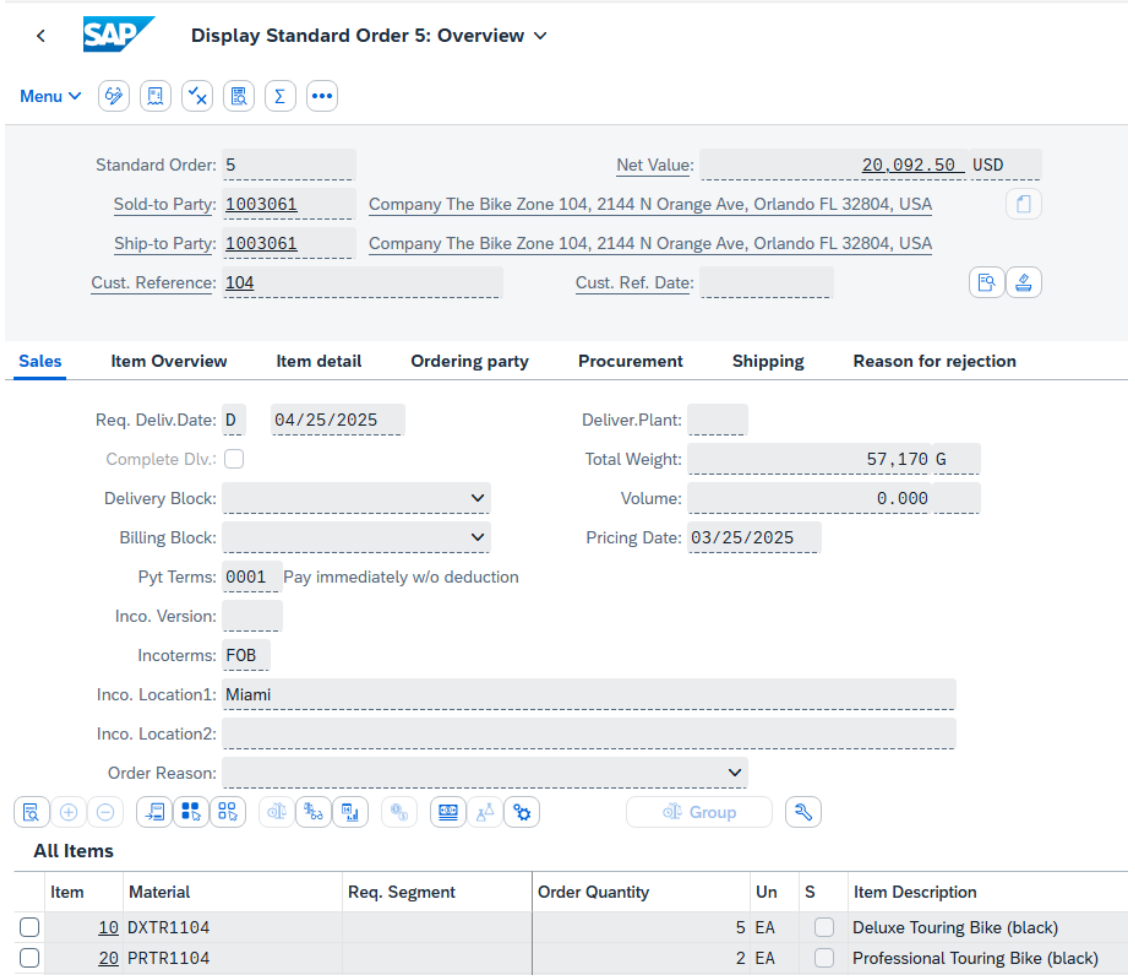
- Sales order processing can originate from a variety of documents and activities
  - Customer contacts us for an order: phone, internet, e-mail
  - Existing Contract
  - Quotations

- The electronic document that is created should contain the following basic information:
  - Customer Information
  - Material/service and quantity
  - Pricing (conditions)
  - Specific delivery dates and quantities
  - Shipping information
  - Billing Information

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Sales Order

- The sales document is made up of three primary areas:
  - Header
    - Data relevant for the entire sales order: e.g.: customer data, total cost of the order
  - Line Item(s)
    - Information about the specific product: e.g.: material and quantity, cost of an individual line
  - Schedule Line(s)
    - Uniquely belongs to a Line Item, contains delivery quantities and dates for partial deliveries



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Sales Order

- The sales order contains all of the information needed to process your customers request

- The following information is determined for each sales order:
  - Delivering Schedule
  - Shipping point and route determination
  - Availability Check
  - Transfer of requirements to MRP
  - Pricing
  - Credit limit check

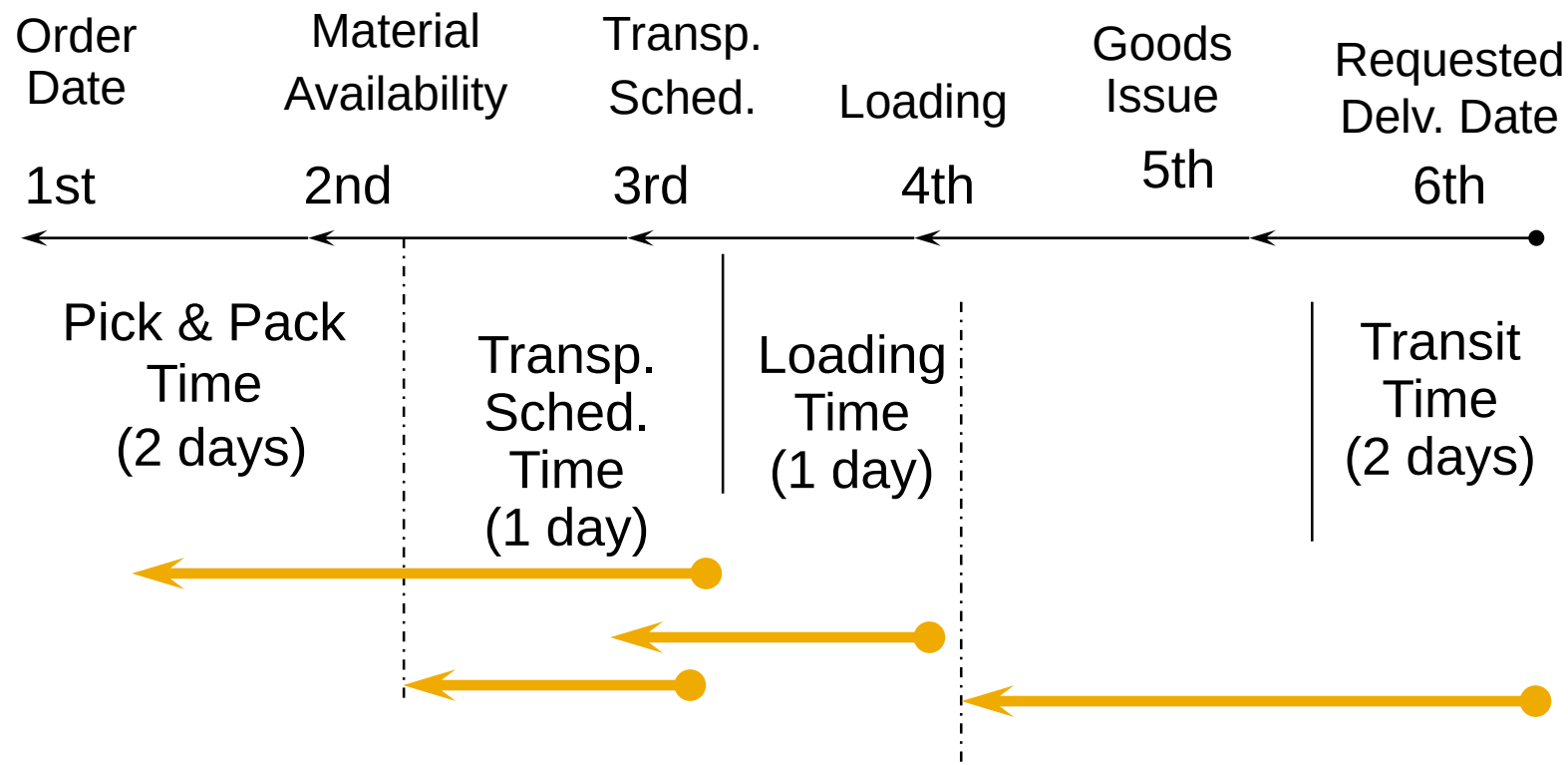
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Delivery Scheduling

- When an order is created you must enter a requested delivery date for the order or each line item.
- The system will then determine a delivery timeline, this will be used when determining our material availability, or ATP (Availability to Promise) date.
- The system will determine this date using forward and backward scheduling rules you have defined.

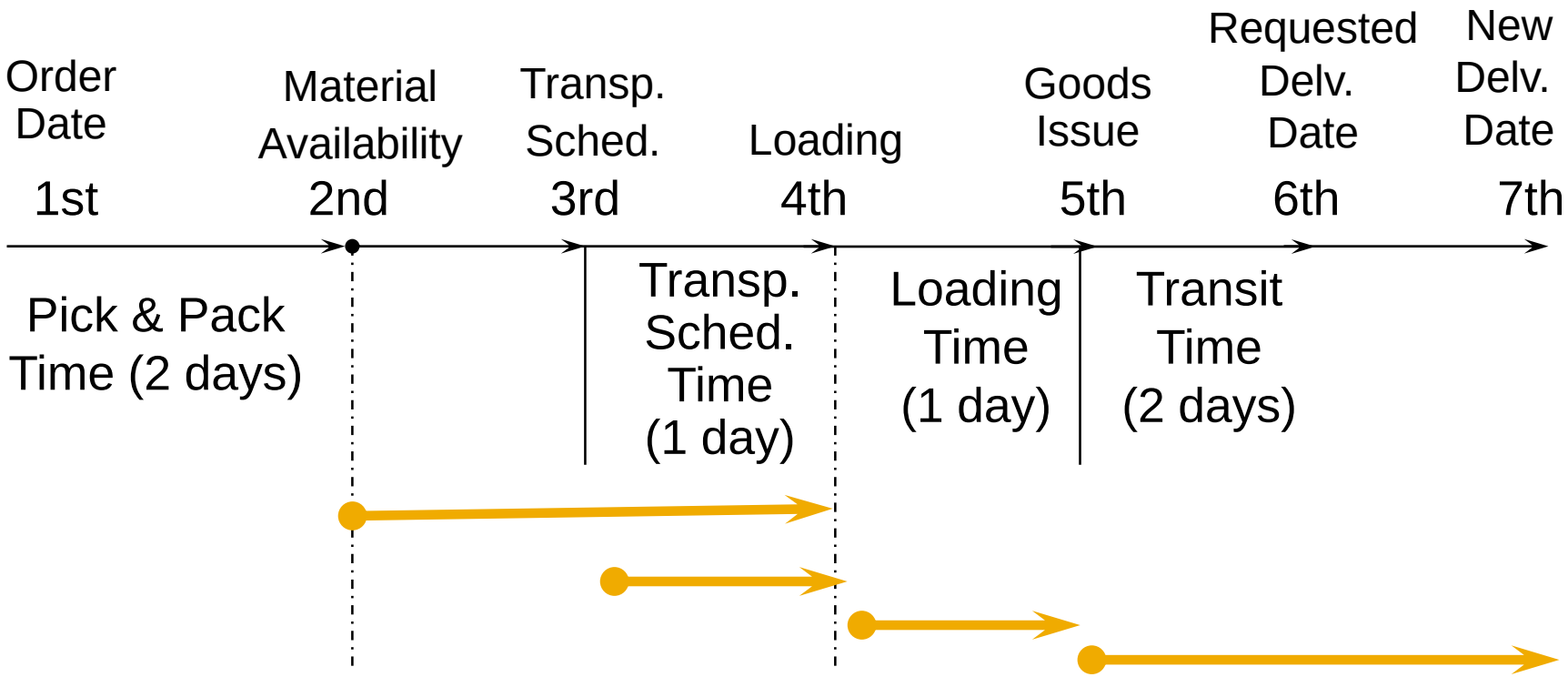
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Backward Scheduling



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Forward Scheduling



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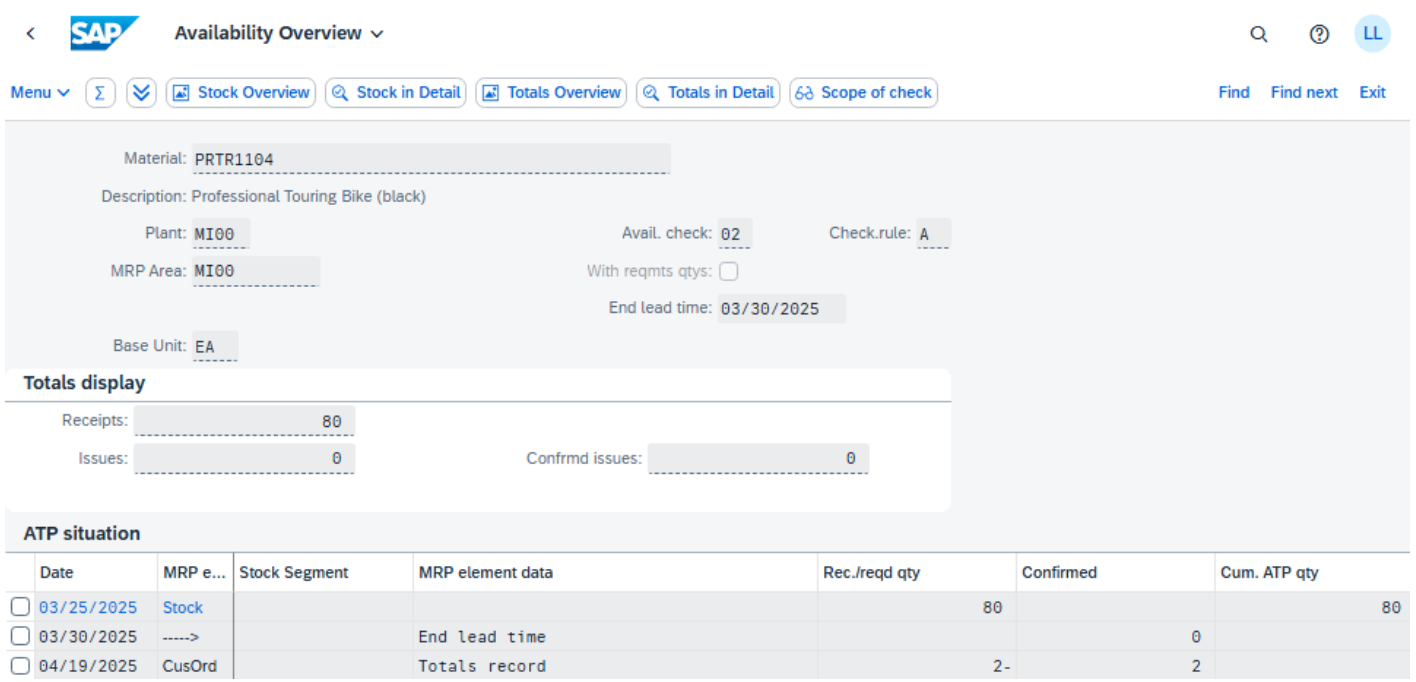
Shipping & Route Determination

- During the creation of the sales order, the system must determine the shipping point from which the material will be shipped and the route the material will take to get from your warehouse to your customers location.
- A shipping point is determined for each line item within the order.
- The route determination is used to define the transit time of the material that we used in scheduling.

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Availability Check

- Availability Check
  - Determines the material availability date
  - Considers all inward and outward inventory movements
- Proposes 3 methods of delivery:
  - One-time delivery
  - Complete delivery
  - Delayed proposal
- Rules are created by USERS



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Pricing

- The system displays pricing information for all sales documents on the pricing screens at both the header and the line item level
  - Header pricing is valid for the whole order and it is the cumulative of all line items within the order
  - Line item pricing is for each specific material.
- The system will automatically search for price, discounts, surcharges, calculated taxes and freight
- You have the ability to manually manipulate the pricing at both the header and line item level within the sales order by entering a condition type and amount
  - Taxes and freight can be set-up automatically so we can not manually enter them

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Credit Check

- Allows your company to manage its credit exposure and risk for each customer by specifying credit limits
- During the sales order process the system will alert the sales report about the customers’ credit situation that arises
- If necessary, the system can be configured to block orders and deliveries

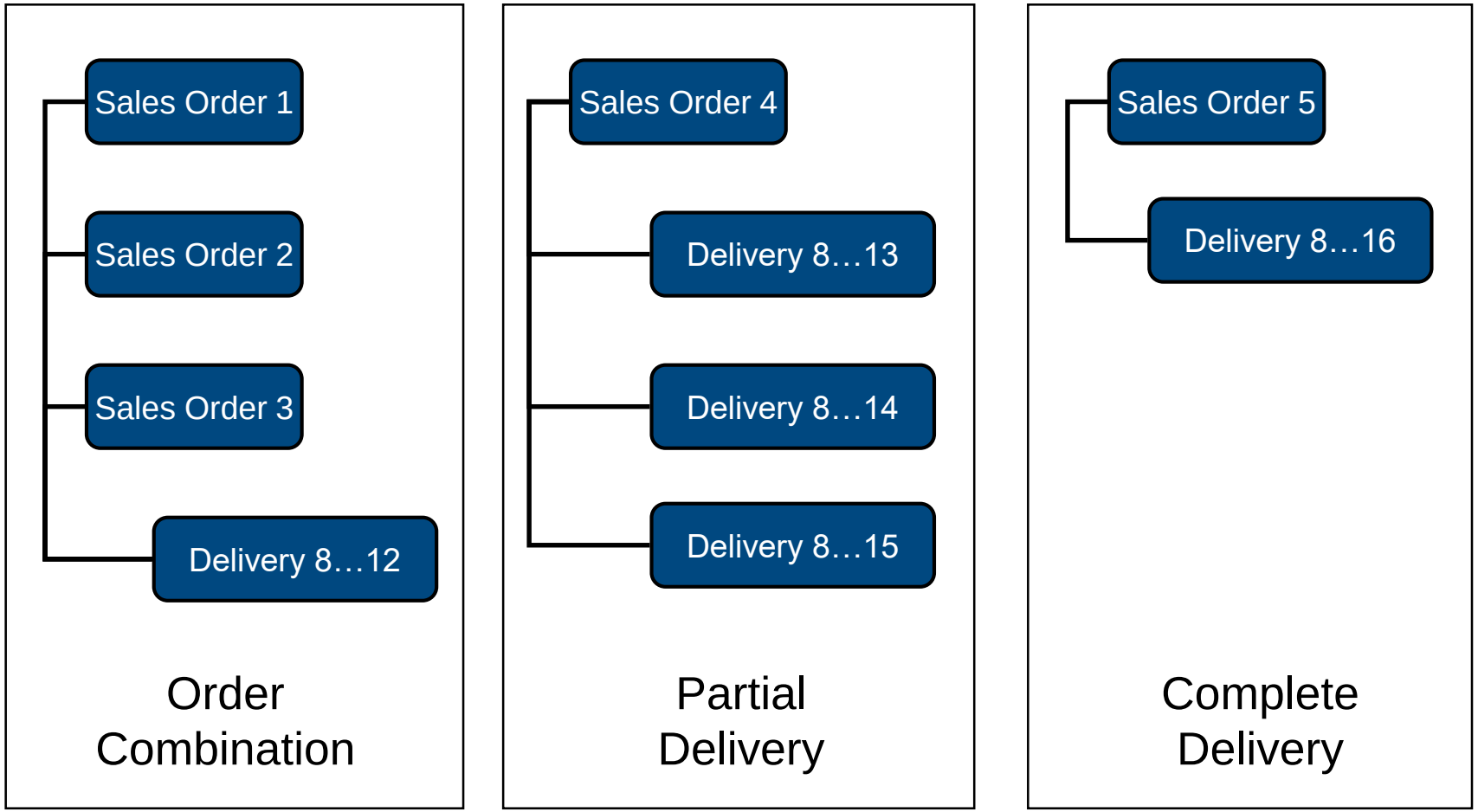
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Shipping & Transportation

- The shipping process begins when you create the delivery document for the sales order. This document controls, supports, and monitors numerous sub-processes for progressing the shipping:
  - Picking
  - Packing
  - Post Goods Issue
- Integrated with the Material Management (MM), extended Warehouse Management (eWM) and Finance (FI) modules

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Shipping



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Delivery Creation

- Checks order and materials to determine if a delivery is possible — delivery block (hold), completeness
- Confirms availability
- Determines total weight & volume
- Generates packing proposal
- Checks scheduling
- Considers partial deliveries
- Updates route assignment
- Assigns picking location
- Updates sales order
- Determines batches
- Quality check (if needed)
- Updates sales order

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Delivery Document

- The Delivery Document initiates the delivery process and is the control mechanism for the following processes:
  - Picking
  - Packing
  - Loading
  - Posting Goods Issue
- Changes to delivery are allowable - products, quantities

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Picking

- Quantities based on delivery note
- Assigned date when picking should begin
- Automated storage location assignment
- Supports serial number/lot number tracking and batch management
- Integrated with Warehouse Management (WM)

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Loading and Packing

- Identifies which packaging is to be used for specified products (customer preference and UCC-128 considerations)
- Identifies and updates accounts associated with returnable packaging
- Tracks the packed product by container
- Insures weight/volume restrictions are enforced
- All packed items are assigned to the required means of transportation

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Goods issue

- Event that could indicates the legal change in ownership of the products
  - It signals that the goods have left the premises of the seller. Legal change in ownership depends on the “Incoterm” and what is written in the sales order terms and conditions.
- Reduces inventory and enters Cost of Goods Sold
- Automatically updates the General Ledger (G/L) accounts
- Ends the shipping process and updates the status of the shipping documents

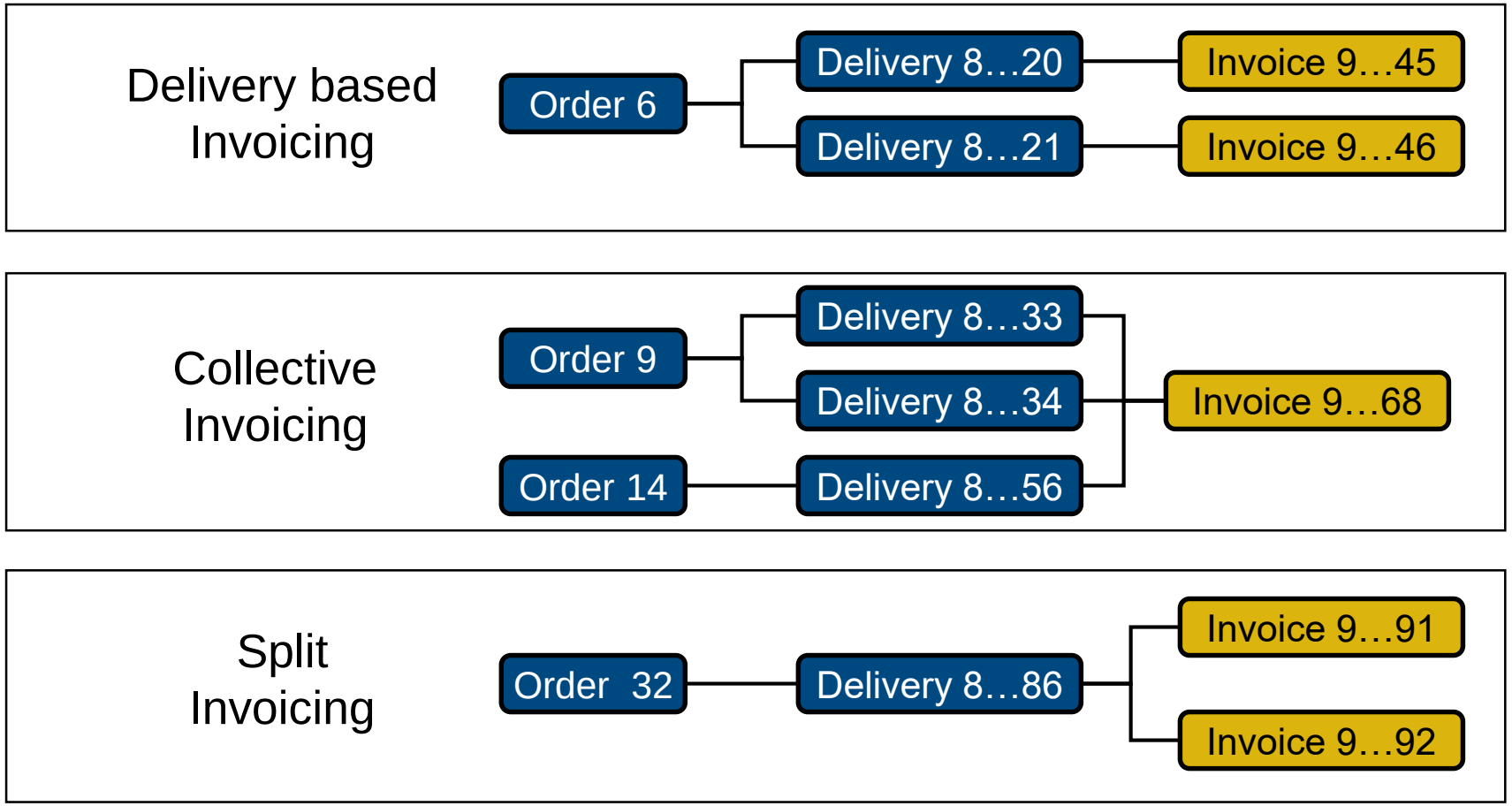
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Billing

- The billing document is created by coping data from the sales order and/or delivery document.
  - Order-based billing
  - Delivery-based billing
- The billing process is used to generate the customer invoice
- It will update the customer’s credit status.
- The billing document will automatically create a debit posting to your customer’s sub-ledger account and credit it to your revenue account
- It is at this point that the sales process is passed over to Financial Accounting to await payment.

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Billing Methods



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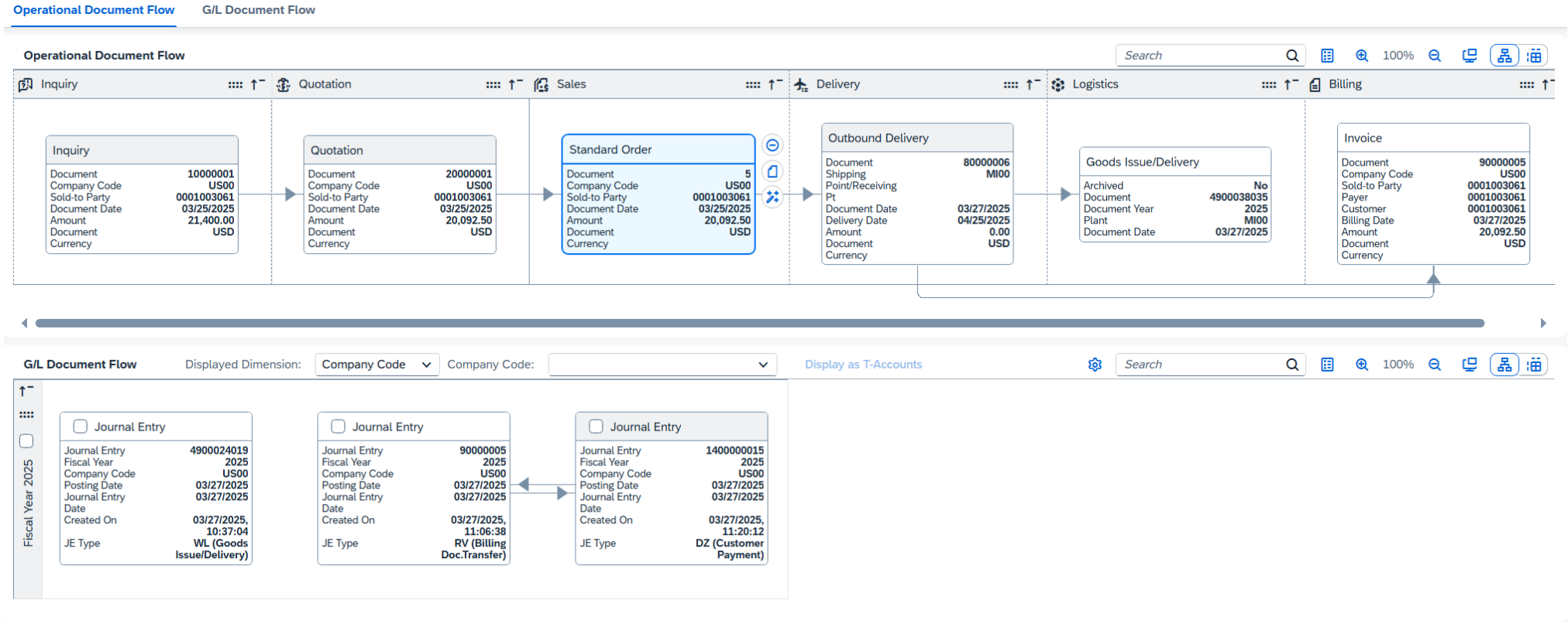
Payment

- Payment is the final step in the sales order process, this step is managed by the Financial Accounting department.
- Final payment includes:
  - Posting payments against invoices
  - Reconciling differences between payment and invoice
- Payment will create a posting clearing the liability in the A/R account and increases your bank account.

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Document Flow

- The document flow and order status feature allows you to find the status of an order at any point in time. The SAP updates the order status every time a change is made to any document created in the customer order management cycle (Order-to-Cash).



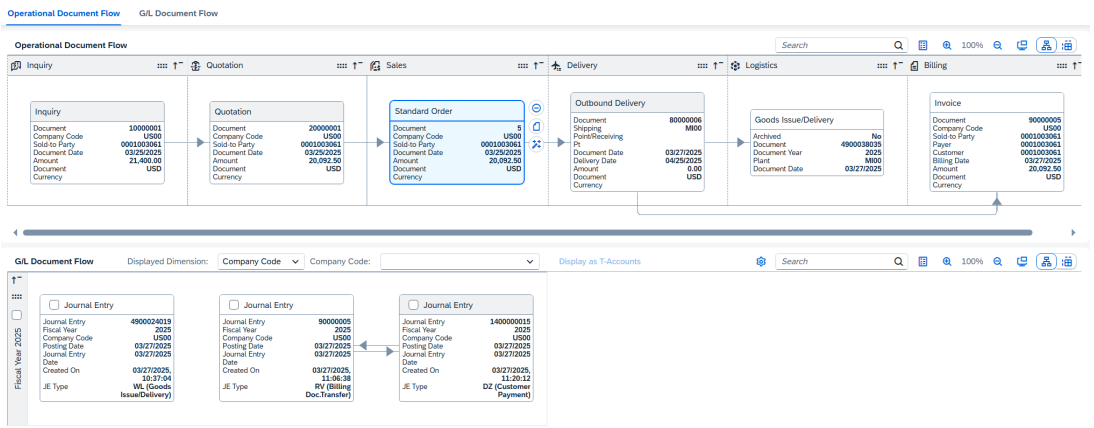
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Sales Order Process Debugging

- “Never again did I learn so much in such a short time, because twenty participants made mistakes for me! You cannot make so many mistakes all alone”

SAP Co-founder Klaus Tschira
on his experiences teaching COBOL
to clients at IBM

- Document Flow
  - Gives Order Process Status
- List of Sales Orders (VA05 / App Manage Sales Orders)
  - Tool to Find Order
  - Need student’s user id/data set number



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